Session 05: Balancing Equations & Mole Ratios

> **Topic:** From a pile of one substance to a pile of another — the mole ratio is the bridge.
> **Time:** 150 minutes
> **Big Fat Notebook:** Chapter 7

Opening Attack

Try this:

> You burn **50.0 g of propane (C_3H_8, "propane")** in excess oxygen.
>
> Propane burns according to this unbalanced equation:
>
> $$C_3H_8 + O_2 \rightarrow CO_2 + H_2O$$
>
> (a) Balance the equation.
>
> (b) How many grams of CO_2 (carbon dioxide) are produced?
>
> (c) How many grams of H_2O (water) are produced?

Stuck? Good. By the end of this session, you'll balance any equation by inspection, extract mole ratios, and convert grams of reactant to grams of product like a machine.

Worked Pattern

Example 1 — Balancing by Inspection: Simple

Problem: Balance: $H_2 + O_2 \rightarrow H_2O$

The inspection method — count atoms on each side of the arrow:

Side	H	O
Left	2	2
Right	2	1

Oxygen is unbalanced: 2 on left, 1 on right.

Fix O: Put a 2 in front of H_2O.

> $H_2 + O_2 \to \mathbf{2}\,H_2O$

Recount:

Side	H	O
Left	2	2
Right	$2 \times 2 = 4$	$2 \times 1 = 2$

Oxygen is balanced ($2 = 2$). Hydrogen is now unbalanced ($2 \neq 4$).

Fix H: Put a 2 in front of H_2.

> $\mathbf{2}\,H_2 + O_2 \to 2\,H_2O$

Final count:

Side	H	O
Left	$2 \times 2 = 4$	2
Right	$2 \times 2 = 4$	$2 \times 1 = 2$

All balanced. ✓

Answer: $\boxed{2H_2 + O_2 \to 2H_2O}$.

$$\underbrace{2H_2}_{\text{2 molecules}} + \underbrace{O_2}_{\text{1 molecule}} \xrightarrow{\text{react}} \underbrace{2H_2O}_{\text{2 molecules}}$$

$$\text{Mole ratio: } H_2 : O_2 : H_2O = 2 : 1 : 2$$

Example 2 — Balancing by Inspection: Moderate

Problem: Balance: $Fe + O_2 \rightarrow Fe_2O_3$

Count atoms:

Side	Fe	O
Left	1	2
Right	2	3

Both are unbalanced.

Start with Fe: Put a 2 in front of Fe.

> $\mathbf{2}\,Fe + O_2 \rightarrow Fe_2O_3$

Side	Fe	O
Left	2	2
Right	2	3

Fe is balanced. O is not ($2 \neq 3$).

To balance O, find the LCM of 2 and 3 = 6.

> Right side: multiply Fe_2O_3 by $2 \rightarrow \mathbf{2}\,Fe_2O_3$ (has 6 O)
>
> Left side: multiply O_2 by $3 \rightarrow \mathbf{3}\,O_2$ (has 6 O)

> $2\,Fe + 3\,O_2 \rightarrow 2\,Fe_2O_3$

Recount Fe: Left = 2, Right = $2 \times 2 = 4$. Unbalanced!

Fix Fe: Put a 4 in front of Fe.

> $4\,Fe + 3\,O_2 \rightarrow 2\,Fe_2O_3$

Final count:

Side	Fe	O
Left	4	$3 \times 2 = 6$
Right	$2 \times 2 = 4$	$2 \times 3 = 6$

All balanced. ✓

Answer: $\boxed{4Fe + 3O_2 \rightarrow 2Fe_2O_3}$.

> **Tip:** Leave the pure element (Fe) for last. Balance the element "trapped" in compounds first.

Example 3 — Balancing a Combustion Reaction

Problem: Balance the combustion of propane: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

Combustion pattern: $C_xH_y + O_2 \rightarrow CO_2 + H_2O$

Always balance in this order: **C** $\rightarrow$ **H** $\rightarrow$ **O**

Step 1 — Balance carbon (C):

> Left: C = 3 (from C_3H_8)
>
> Right: each CO_2 has 1 C $\rightarrow$ need **3** CO_2.

> $C_3H_8 + O_2 \rightarrow \mathbf{3}\, CO_2 + H_2O$

Step 2 — Balance hydrogen (H):

> Left: H = 8
>
> Right: each H_2O has 2 H → need **4** H_2O.
>
> $C_3H_8 + O_2 \rightarrow 3\, CO_2 + \mathbf{4}\, H_2O$

Step 3 — Balance oxygen (O):

> Right: O from $3\, CO_2 = 3 \times 2 = 6$, O from $4\, H_2O = 4 \times 1 = 4$
>
> Total O on right = $6 + 4 = 10$
>
> Left: each O_2 has 2 O → need **5** O_2.
>
> $C_3H_8 + \mathbf{5}\, O_2 \rightarrow 3\, CO_2 + 4\, H_2O$

Final count:

Side	C	H	O
Left	3	8	$5 \times 2 = 10$
Right	3	$4 \times 2 = 8$	$3 \times 2 + 4 \times 1 = 10$

All balanced. ✓

Answer: $\boxed{C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O}$.

$$\begin{aligned} \text{Mole ratios:} \quad C_3H_8 : O_2 &= 1 : 5 \quad (\text{1 mol propane needs 5 mol } O_2) \\ C_3H_8 : CO_2 &= 1 : 3 \quad (\text{1 mol propane makes 3 mol } CO_2) \\ C_3H_8 : H_2O &= 1 : 4 \quad (\text{1 mol propane makes 4 mol } H_2O) \end{aligned}$$

Example 4 — Reading Mole Ratios from a Balanced Equation

Problem: Given $N_2 + 3H_2 \rightarrow 2NH_3$, how many moles of NH_3 (ammonia) are produced from 4.00 mol of N_2 (nitrogen gas)?

Step 1: Identify the mole ratio from the balanced equation.

> $N_2 : NH_3 = 1 : 3$? No — wait! Check the coefficients:
>
> N_2 coefficient = **1**, NH_3 coefficient = **2**.
>
> Ratio = $1 : 2$, **not** $1 : 3$!
>
> Careful: the 3 is on H_2, not NH_3.

Step 2: Use the ratio to convert.

> $\text{mol } NH_3 = 4.00 \text{ mol } N_2 \times \dfrac{2 \text{ mol } NH_3}{1 \text{ mol } N_2} = 8.00 \text{ mol } NH_3$

Step 3: Verify: the units cancel.

> $\cancel{\text{mol } N_2} \times \dfrac{\text{mol } NH_3}{\cancel{\text{mol } N_2}} = \text{mol } NH_3. \checkmark$

Answer: $\boxed{8.00 \text{ mol } NH_3}$.

$$\text{Mole ratio} = \frac{\text{coefficient of target}}{\text{coefficient of given}} = \frac{2}{1}$$

$$\therefore \text{ The mole ratio is always from COEFFICIENTS, never from subscripts.}$$

Example 5 — Mass–Mass Stoichiometry: The Full Chain

Problem: How many grams of CO_2 are produced when 50.0 g of C_3H_8 burns? (Use the balanced equation from Example 3.)

The balanced equation: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

Step 1: Convert grams of given to moles.

> Molar mass $C_3H_8 = 3(12.01) + 8(1.008) = 44.094 \text{ g/mol}$
>
> $\text{mol } C_3H_8 = 50.0 \text{ g} \div 44.094 \text{ g/mol} = 1.1339 \text{ mol } C_3H_8$

Step 2: Use the mole ratio to find moles of target.

> From equation: $1 \text{ mol } C_3H_8 \rightarrow 3 \text{ mol } CO_2$
>
> $\text{mol } CO_2 = 1.1339 \text{ mol } C_3H_8 \times \frac{3 \text{ mol } CO_2}{1 \text{ mol } C_3H_8} = 3.4017 \text{ mol } CO_2$

Step 3: Convert moles of target to grams.

> Molar mass $CO_2 = 12.01 + 2(16.00) = 44.01 \text{ g/mol}$
>
> $\text{g } CO_2 = 3.4017 \text{ mol} \times 44.01 \text{ g/mol} = 149.71 \text{ g}$
>
> Sig figs: 50.0 (3 sig figs) $\rightarrow 150.$ g CO_2

Answer: $\boxed{150. \text{ g } CO_2}$ (or 1.50×10^2 g).

$$\underbrace{50.0 \text{ g } C_3H_8}_{\text{grams given}} \xrightarrow{\div 44.094} \underbrace{1.134 \text{ mol } C_3H_8}_{\text{moles given}} \xrightarrow{\times \frac{3}{1}} \underbrace{3.402 \text{ mol } CO_2}_{\text{moles target}} \xrightarrow{\times 44.01} \underbrace{150. \text{ g } CO_2}_{\text{grams target}}$$

Three steps. One answer. Every mass–mass stoichiometry problem follows this chain.

⚠️ THE TRAP — Using the Mole Ratio Before Converting to Moles

The wrong way:

> How many grams of H_2O are produced from 50.0 g of C_3H_8?

> *Wrong work:* "The ratio is 1 : 4, so 50.0 × 4 = $200 \text{ g of } H_2O$." ← **WRONG!**

Why is this wrong? The mole ratio $(1:4)$ relates **moles**, not grams. You cannot multiply grams by the mole ratio — the units don't match. You must convert grams to moles FIRST, then apply the ratio, then convert back to grams.

The right way:

Step 1: $\text{mol } C_3H_8 = 50.0 \text{ g} \div 44.094 \text{ g/mol} = 1.1339 \text{ mol } C_3H_8$

Step 2: $\text{mol } H_2O = 1.1339 \text{ mol } C_3H_8 \times \dfrac{4 \text{ mol } H_2O}{1 \text{ mol } C_3H_8} = 4.5356 \text{ mol } H_2O$

Step 3: $\text{g } H_2O = 4.5356 \text{ mol} \times 18.016 \text{ g/mol} = 81.71 \text{ g}$

> Sig figs: 50.0 (3 sig figs) $\rightarrow 81.7 \text{ g } H_2O$

Answer: $\boxed{81.7 \text{ g } H_2O}$.

$$\text{Spot the difference: } 200 \text{ g (wrong) vs. } 81.7 \text{ g (right).}$$

The wrong method gives over **twice** the correct answer — and on an exam, the wrong answer is sitting right there as a distractor.

$$\boxed{\text{THE GOLDEN RULE OF STOICHIOMETRY}}$$

$$\text{Grams of A} \xrightarrow{\div M_A} \text{Moles of A} \xrightarrow{\times \text{ mole ratio}} \text{Moles of B} \xrightarrow{\times M_B} \text{Grams of B}$$

> Never jump directly from grams of A to grams of B. Always pass through the mole tunnel.

Example 6 — Mass–Mass with a Different Given

Problem: Using $2Al + 3Cl_2 \rightarrow 2AlCl_3$, how many grams of Cl_2 (chlorine gas) are needed to react completely with 25.0 g of Al (aluminum)?

Step 1: $\text{g Al} \rightarrow \text{mol Al}$.

> Molar mass Al = 26.98 g/mol
>
> $$\text{mol Al} = 25.0 \text{ g} \div 26.98 \text{ g/mol} = 0.92661 \text{ mol Al}$$

Step 2: $\text{mol Al} \to \text{mol } Cl_2$ (mole ratio).

> From equation: $2 \text{ Al} : 3\ Cl_2$ $\rightarrow$ ratio = $\dfrac{3}{2}$
>
> $$\text{mol } Cl_2 = 0.92661 \text{ mol Al} \times \frac{3 \text{ mol } Cl_2}{2 \text{ mol Al}} = 1.3899 \text{ mol } Cl_2$$

Step 3: $\text{mol } Cl_2 \to \text{g } Cl_2$.

> Molar mass $Cl_2 = 2 \times 35.45 = 70.90 \text{ g/mol}$
>
> $$\text{g } Cl_2 = 1.3899 \text{ mol} \times 70.90 \text{ g/mol} = 98.54 \text{ g}$$
>
> Sig figs: 25.0 (3 sig figs) $\rightarrow 98.5 \text{ g } Cl_2$

Answer: $\boxed{98.5 \text{ g } Cl_2}$.

$$\cancel{\text{mol Al}} \times \frac{\text{mol } Cl_2}{\cancel{\text{mol Al}}} = \text{mol } Cl_2 \checkmark$$

Example 7 — Multi-Step: Both Products

Problem: Using $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$, 50.0 g of propane produces CO_2 and H_2O. Find the grams of BOTH products.

CO_2 path (from Example 5):

> $$\text{mol } C_3H_8 = 50.0 \div 44.094 = 1.1339 \text{ mol}$$
>
> $$\text{mol } CO_2 = 1.1339 \times 3 = 3.4017 \text{ mol}$$
>
> $$\text{g } CO_2 = 3.4017 \times 44.01 = 149.7 \text{ g}$$

H_2O **path:**

> $\text{mol } H_2O = 1.1339 \times 4 = 4.5356 \text{ mol}$
>
> $\text{g } H_2O = 4.5356 \times 18.016 = 81.71 \text{ g}$

Answers: $\boxed{150. \text{ g } CO_2, \ 81.7 \text{ g } H_2O}$.

$$\begin{aligned} CO_2 + H_2O &= 149.7 \text{ g} + 81.71 \text{ g} \\ &= 231.4 \text{ g total products} \end{aligned}$$

Example 8 — Stoichiometry with a Trick: "Excess" Reagent

Problem: 50.0 g of C_3H_8 burns in **excess** oxygen. How many grams of O_2 are actually consumed?

When a reagent is "in excess," the other reagent (C_3H_8) determines how much reaction occurs. We calculate from the non-excess reagent.

Step 1: $\text{mol } C_3H_8 = 50.0 \text{ g} \div 44.094 \text{ g/mol} = 1.1339 \text{ mol}$

Step 2: $\text{mol } O_2 = 1.1339 \text{ mol } C_3H_8 \times \dfrac{5 \text{ mol } O_2}{1 \text{ mol } C_3H_8} = 5.6695 \text{ mol } O_2$

Step 3: $\text{g } O_2 = 5.6695 \text{ mol} \times 32.00 \text{ g/mol} = 181.42 \text{ g}$

> Sig figs: 50.0 (3 sig figs) $\rightarrow 181 \text{ g } O_2$

Answer: $\boxed{181 \text{ g } O_2 \text{ consumed}}$.

$$\text{"Excess"} \implies \text{calculate based on the other reagent.}$$

Session 06 will teach you what to do when NEITHER is in excess — and you must find the limiting reagent.

What We Just Did — The Mass–Mass Stoichiometry Procedure

Step 1: Convert grams of the given substance to moles.

> $\text{mol} = \text{grams} \div \text{molar mass (g/mol)}$
>
> You MUST know the molar mass of the given substance.
>
> This step is NOT optional. You cannot skip to the mole ratio.

Step 2: Use the mole ratio from the balanced equation.

> $\text{mol target} = \text{mol given} \times \dfrac{\text{coefficient of target}}{\text{coefficient of given}}$
>
> The coefficients come from the BALANCED equation, not subscripts.
>
> If the equation isn't balanced yet, balance it FIRST.

Step 3: Convert moles of the target substance to grams.

> $\text{grams} = \text{mol} \times \text{molar mass of target (g/mol)}$
>
> Apply sig figs only at THIS final step.

$$\text{g A} \xrightarrow{\div M_A} \text{mol A} \xrightarrow{\times \frac{\text{coeff B}}{\text{coeff A}}} \text{mol B} \xrightarrow{\times M_B} \text{g B}$$

$$\boxed{\text{Never skip the mole step. Never mix up grams and moles.}}$$

Copy this. Every mass–mass stoichiometry problem follows these three steps. The numbers change; the chain does not.

💡 Pro Tips — Calculator Shortcuts & Mental Checks

Tip 1 — Combustion balancing order is ALWAYS C → H → O. Never start with oxygen in a combustion reaction — you'll chase your tail. Balance carbon first, then hydrogen, then count oxygen on the right and divide by 2.

Tip 2 — The chain never changes. $\text{g A} \rightarrow \text{mol A} \rightarrow \text{mol B} \rightarrow \text{g B}$. If you find yourself multiplying grams directly by a mole ratio, **stop** — you skipped a step. Go back to moles.

Tip 3 — LCM for O_2. When oxygen appears as O_2 on one side and in multiple compounds on the other, find the LCM of the oxygen counts. Example: right side has 3 O (from Fe_2O_3) → left needs O_2. LCM of 2 and 3 is 6 → use 3 O_2 and 2 Fe_2O_3.

Tip 4 — Mole ratio vs. mass ratio. Mole ratio ≠ mass ratio. In $2H_2 + O_2 \rightarrow 2H_2O$, the mole ratio $H_2 : O_2$ is 2:1, but the mass ratio is $2(2.02) : 32.00 = 4.04 : 32.00 \approx 1 : 8$. You need ~8× more O_2 by mass — don't let this surprise you on an exam.

Tip 5 — Quick product check. If your calculated product mass is less than your starting mass in a combustion reaction, something is wrong — you're adding oxygen from the air. 50 g propane + 181 g O_2 → 231 g products. Products should weigh **more** than the fuel alone.

Basic Drills (B1–B10)

> Goal: Automate balancing and mass–mass conversion.

B1. Balance: $Na + Cl_2 \rightarrow NaCl$. Then: how many moles of $NaCl$ are produced from 3.00 mol of Na?

B2. Balance: $CH_4 + O_2 \rightarrow CO_2 + H_2O$. Then: how many moles of O_2 are needed to burn 2.00 mol of CH_4?

B3. Balance: $Mg + O_2 \rightarrow MgO$. Then: how many grams of MgO (magnesium oxide) are produced from 12.0 g of Mg (magnesium)? (Mg = 24.31, O = 16.00 g/mol)

B4. Balance: $KClO_3 \rightarrow KCl + O_2$. Then: how many grams of O_2 are produced from 50.0 g of $KClO_3$? (K = 39.10, Cl = 35.45, O = 16.00 g/mol)

B5. Given $2H_2 + O_2 \rightarrow 2H_2O$: How many grams of H_2O are produced from 10.0 g of H_2? (H = 1.008, O = 16.00 g/mol)

B6. Balance: $Al + HCl \rightarrow AlCl_3 + H_2$. Then: how many grams of H_2 (hydrogen gas) are produced from 15.0 g of Al? (Al = 26.98, H = 1.008, Cl = 35.45 g/mol)

B7. (Word problem) Iron reacts with oxygen to form iron(III) oxide: $Fe + O_2 \rightarrow Fe_2O_3$. Balance the equation. Then calculate the mass of Fe_2O_3 produced from 25.0 g of Fe. (Fe = 55.85, O = 16.00 g/mol)

B8. (Combustion pattern) Balance: $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$. Then: how many grams of CO_2 are produced from 30.0 g of C_2H_6? (C = 12.01, H = 1.008, O = 16.00 g/mol)

B9. (Two-part) Balance: $N_2 + H_2 \rightarrow NH_3$. Part 1: How many grams of NH_3 (ammonia) are produced from 28.0 g of N_2? Part 2: How many grams of H_2 are consumed? (N = 14.01, H = 1.008 g/mol)

B10. (Speed round — under 2 min) Balance: $C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$. Then: 20.0 g of butane (C_4H_{10}) burns. Find grams of CO_2 produced. Use C = 12.01, H = 1.008, O = 16.00.

Advanced Drills (A1–A10)

> Goal: Exam-level multi-step stoichiometry with traps and time pressure.

A1. (Multi-step with Session 04) Balance: $CaCO_3 \rightarrow CaO + CO_2$. (a) How many grams of CaO (quicklime) are produced from 100.0 g of $CaCO_3$ (limestone)? (b) How many CaO formula units is this? (Ca = 40.08, C = 12.01, O = 16.00)

A2. (Multi-step with Session 03) A compound is 85.6% C and 14.4% H by mass. (a) Find the empirical formula. (b) Write the balanced combustion equation for this compound. (c) How many grams of CO_2 are produced from burning 25.0 g of this compound?

A3. (Trap-laden) A student balances $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$ as $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ — which is correct. Then they calculate: "50.0 g C_3H_8 → 1.134 mol → times mole ratio 3 → 3.402 mol CO_2. Now, instead of multiplying by 44.01 g/mol, I'll multiply by 12.01

(just the carbon). 3.402 × 12.01 = 40.9 g C." What two mistakes did the student make? Correct the calculation.

A4. (Reverse engineering) A student calculates that burning 10.0 g of CH_4 produces 55.0 g of CO_2. Their partner gets 27.4 g. (a) Which answer is correct? (b) What mistake did the wrong student make? (c) Show the full correct calculation with the balanced equation.

A5. (Constructive) A classic exam problem: "Nitrogen gas reacts with hydrogen gas to form ammonia." (a) Write and balance the equation. (b) Write a problem where the student must find grams of NH_3 from 50.0 g of N_2. (c) Solve it. (d) Add a second part: find the mass of H_2 consumed.

A6. (Constructive) Write a problem where a student must balance a combustion reaction of an alcohol (e.g., C_2H_5OH) and then calculate both products' masses from a given mass of alcohol. Solve it. Include a common pitfall in the solution notes.

A7. (Table-based) Four reactions and their balanced equations:

Reaction	Balanced Equation
1	$2Mg + O_2 \rightarrow 2MgO$
2	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
3	$4Fe + 3O_2 \rightarrow 2Fe_2O_3$
4	$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$

For each reaction, 100.0 g of the first reactant is consumed. (a) Calculate grams of each oxygen-containing product. (b) Which reaction consumes the most grams of O_2? (c) Which reaction produces the most grams of CO_2? Use: Mg = 24.31, Fe = 55.85, C = 12.01, H = 1.008, O = 16.00.

A8. (Timed — under 3 min) The thermite reaction: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$. (a) How many grams of molten iron are produced from 50.0 g of Fe_2O_3 and excess Al? (b) How many grams of Al_2O_3? (Fe = 55.85, Al = 26.98, O = 16.00)

A9. (Hardest variation) A 10.00 g sample of a mixture of $KClO_3$ and KCl is heated. The $KClO_3$ decomposes: $2KClO_3 \rightarrow 2KCl + 3O_2$. After heating, the mass of the remaining solid is 8.50 g. The mass lost is all O_2 gas. (a) Calculate the mass of O_2 lost. (b) Calculate the moles of O_2. (c) Use stoichiometry to find the mass of $KClO_3$ in the original mixture. (d) What percent of the original mixture was $KClO_3$? (K = 39.10, Cl = 35.45, O = 16.00)

A10. (Exam boss) A 5.000 g sample of an unknown hydrocarbon (C_xH_y) is burned in excess oxygen, producing 15.69 g of CO_2 and 6.42 g of H_2O. (a) Find the mass of C in the original sample. (b) Find the mass of H in the original sample. (c) Determine the empirical formula of the hydrocarbon. (d) The molar mass of the hydrocarbon is 56.11 g/mol. Find the molecular formula. (e) Write the balanced combustion equation for this hydrocarbon. (f) Verify that the combustion of 5.000 g of this compound produces the given masses of CO_2 and H_2O. Use correct sig figs throughout. (Answer: C_4H_8, butene)

Now Read

You can now balance combustion reactions by inspection, extract mole ratios, and run the full gram → mole → mole → gram chain without hesitation. 50.0 g of propane → 150. g CO_2 + 81.7 g H_2O. You proved it with numbers.

Read **Chapter 7** of *Everything You Need to Ace Chemistry in One Big Fat Notebook*.

You've solved 20 problems — from balancing $H_2 + O_2$ to reverse-engineering a hydrocarbon's molecular formula from combustion data. The concepts — what a balanced equation actually says about atom counts, why mole ratios work, how stoichiometry connects the lab to the equation — will lock into place.

> Solutions: solutions/05-solutions.md

📝 Key Terms — Quick Reference

Term	What it means
stoichiometry	the quantitative relationship between reactants and products
coefficient	the big number in front of a formula in a balanced equation

Term	What it means
subscript	the small number within a formula (H_2O has subscript 2 on H) — never change these when balancing
mole ratio	$\frac{\text{coeff of target}}{\text{coeff of given}}$ — used to convert between substances
combustion	reaction with O_2 producing CO_2 and H_2O; balance order: **C → H → O**
balancing by inspection	systematic trial: count atoms, adjust coefficients, recount
LCM trick	for O_2 balancing: find LCM of O counts, divide by 2 for the O_2 coefficient
excess	"more than enough" — calculate from the other reagent
g → mol → mol → g	the unbreakable chain — never skip the mole step
mass ratio ≠ mole ratio	$2H_2 + O_2$: mole ratio 2:1, mass ratio ~1:8 — don't confuse them